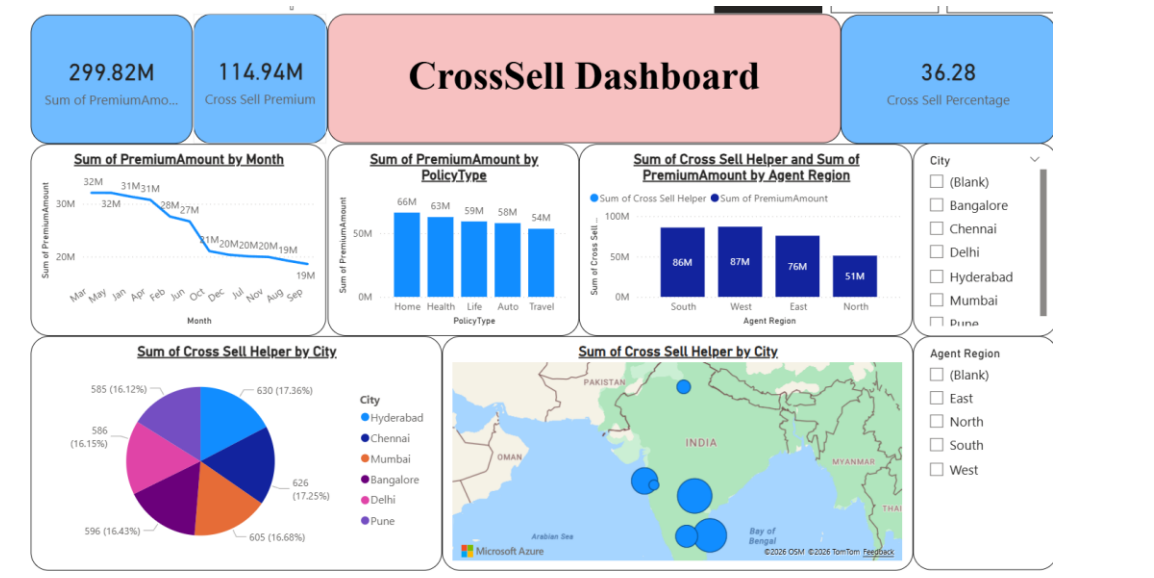


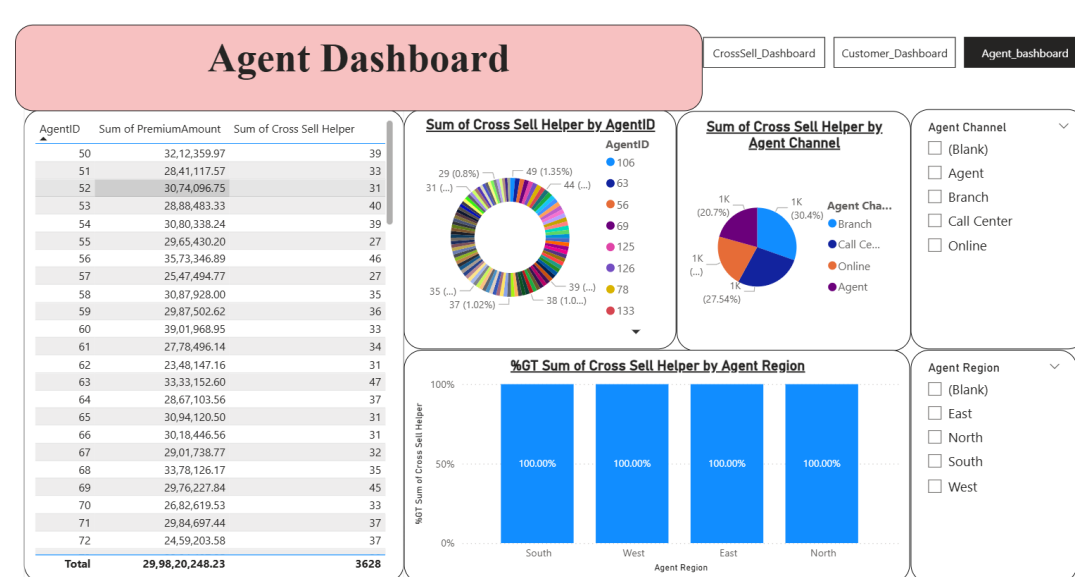
INSURANCE CROSS-SELL ANALYTICS

Using Excel & Power BI

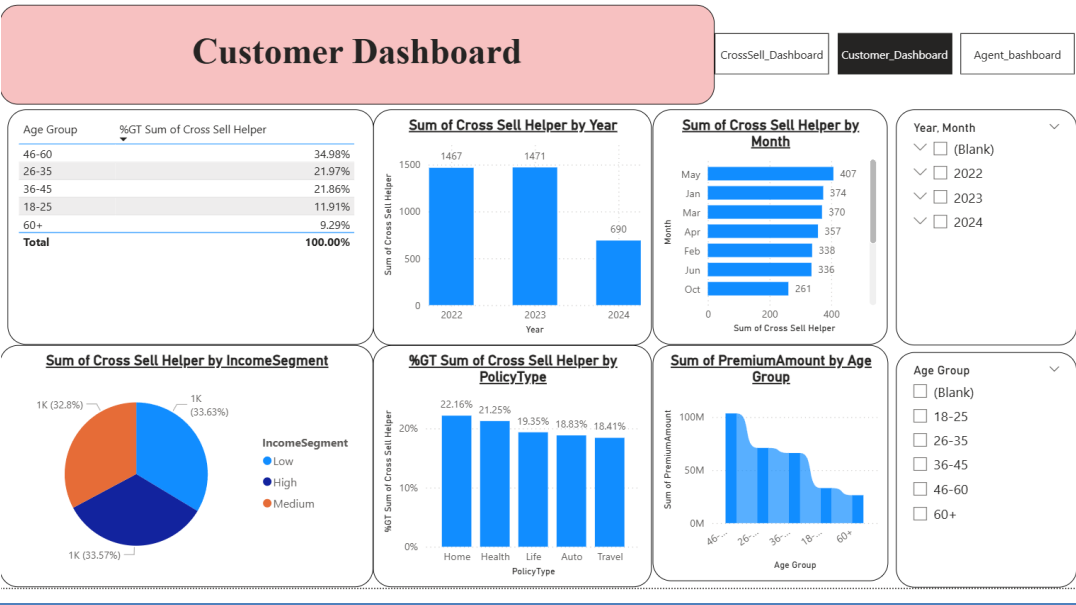
Consolidated Dashboard depicting Single view of Insurance CrossSell comparing various aspects of the newly received insurance premiums.



Consolidated Dashboard depicting Single view of Insurance CrossSell comparing various aspects of the newly received insurance premiums from different agents customers of different sectors



Dashboard showing a single view of Insurance CrossSell comparing various aspects of the newly received insurance premiums from different customers of different sectors



Sum of Cross Sell Helper by IncomeSegment

IncomeSegment	Sum of Cross Sell Helper
Low	1K (32.8%)
High	1K (33.63%)
Medium	1K (33.57%)

%GT Sum of Cross Sell Helper by PolicyType

PolicyType	%GT Sum of Cross Sell Helper
Home	22.16%
Health	21.25%
Life	19.35%
Auto	18.83%
Travel	18.41%

Sum of PremiumAmount by Age Group

Age Group	Sum of PremiumAmount
18+	100M
25+	75M
35+	65M
45+	55M
60+	45M

Age Group

☐ (Blank)

☐ 18-25

☐ 26-35

☐ 36-45

☐ 46-60

☐ 60+

Introduction

In the insurance industry, acquiring new customers is expensive and difficult, making cross-selling to existing customers a key revenue strategy.

This project analyzes insurance sales data to evaluate cross-sell effectiveness using Excel and Power BI.

The primary goal of this project is to analyse customer purchase patterns to drive **cross-selling**—the practice of selling additional products to existing customers. By understanding which customers own specific insurance types (e.g., Home) but lack others (e.g., Life), the business can target marketing efforts more effectively, increase the "Policies per Customer" ratio, and improve overall Customer Lifetime Value (CLV).

We will also be analysing performance of Agents and Channels in cross selling. Analyzing these metrics is critical because cross-selling is rarely passive; it requires an active nudge from a salesperson or a specific marketing touchpoint.

Business Problem Statement

"The organization currently relies on static reporting, leading to delayed decision-making and a lack of visibility into cross-sell performance, customer behaviour and sales channel effectiveness. Static reports delayed decisions and made it difficult to identify growth opportunities.

The organization's current reliance on manual, static reporting has created a "data-lag," where insights are often outdated by the time they reach decision-makers. This lack of real-time visibility has led to:

- **Information Asymmetry:** Stakeholders cannot clearly see the correlation between customer demographics and policy purchase behaviour.
- **Missed Revenue Opportunities:** High-propensity cross-sell targets remain unidentified within the existing customer base.
- **Channel Inefficiency:** An inability to distinguish between high-performing and underperforming sales channels (Agent, Online, Call Centre) in real-time.

Project Aim

To analyze insurance policy data and evaluate cross-sell effectiveness, identify high-performing customers, policies, and channels, and present insights through interactive Power BI dashboards.

Mission: To engineer a robust, interactive Business Intelligence (BI) ecosystem that replaces latent reporting with a single source of truth. This ecosystem will empower the organization to transition from "**What happened?**" to "**What is happening, and where can we grow?**"

Core Objectives:

- **Dynamic Portfolio Visibility:** Provide a 360-degree view of the customer journey, identifying which segments (e.g., High-Income individuals in specific cities) are currently underserved.
- **Performance Optimization:** Implement real-time tracking of Agent and Channel effectiveness to optimize sales force ROI.
- **Growth Identification:** Utilize "White Space Analysis" to automatically highlight customers who hold single policies but fit the profile for additional insurance products.

Project Objectives

Expected Business Outcomes

Leveraging the provided datasets (Customers, Policies, Agents, Transactions, and Payments), the project will follow a sophisticated data pipeline:

1. **Data Consolidation (Excel/Power Query):** Streamlining the raw multi-sheet data into a structured format, ensuring data integrity and cleaning anomalies in transaction records.
 2. **Advanced Data Modelling (Power BI):** Implementing a **Star Schema** model. By linking the Policy Transactions to various dimension tables, we create a multidimensional environment that supports complex filtering.
 3. **Real-Time Analytics Layer:** Moving away from static snapshots to a system where slicers for Region, Income Segment, and Policy Type provide instant answers to ad-hoc business queries.
- Clean and prepare multi-sheet insurance data
 - Design a star schema data model
 - Create KPIs using DAX

- Analyze customer, policy, and channel performance
- Build interactive dashboards

Project Scope

- **Reduced Time-to-Insight:** Eliminating the manual preparation time associated with static reports, allowing management to act on market trends immediately.
- **Enhanced Cross-Sell Conversion:** A projected increase in "Policies per Customer" through data-driven targeting.
- **Proactive Retention:** Identifying "Lapsed" or "Cancelled" trends early enough to implement customer win-back strategies.

In Scope:

- Data cleaning and transformation
- Descriptive analytics
- Dashboard development

Out of Scope:

- Predictive modeling
- Real-time data integration

Dataset Overview

The dataset is a multi-sheet Excel workbook (~10,000 records) containing policy transactions, customers, policies, agents, and payments.

Policy Transactions:

This is the **heart of the analysis**. It records every instance of a policy being sold or updated.

- **PolicyTransactionID:** Unique record for each sale event.
- **TransactionDate:** When the sale occurred (crucial for time-series and trend analysis).
- **CustomerID / PolicyID / AgentID:** Foreign keys that link this transaction to the specific customer, product, and salesperson.
- **PremiumAmount:** The revenue generated. This allows you to calculate the financial impact of cross-selling.
- **PolicyStatus:** Current state of the policy (**Active, Lapsed, Cancelled**). Cross-selling should ideally target "Active" customers.
- **CrossSellFlag (Yes/No):** A direct indicator of whether the transaction was a cross-sell. This is your primary metric for calculating conversion rates.

PolicyTransactionID	TransactionDate	CustomerID	PolicyID	AgentID	PremiumAmount	PolicyStatus	CrossSellFlag
1	2023-06-05 00:00:00	3661	491	95	22730.74	Lapsed	No
2	2024-03-03 00:00:00	4100	546	57	38343.47	Active	No
3	2024-04-15 00:00:00	2830	794	127	58161.51	Active	No
4	2024-05-21 00:00:00	3865	223	118	41132.6	Active	No
5	2024-05-05 00:00:00	3141	486	93	40539.86	Active	Yes
6	2022-03-21 00:00:00	1549	765	114	29286.8	Active	Yes
7	2022-12-17 00:00:00	3950	753	139	11498.24	Active	Yes
8	2024-05-16 00:00:00	2238	705	122	16857.51	Lapsed	Yes
9	2022-02-23 00:00:00	3157	629	67	14215.99	Lapsed	No
10	2023-08-02 00:00:00	4367	256	79	24054.32	Active	No
11	2024-01-09 00:00:00	3459	753	136	27463.44	Active	No
12	2023-06-11 00:00:00	1070	597	101	14421.87	Active	No
13	2024-03-13 00:00:00	1733	596	71	54038.01	Cancelled	No
14	2024-01-23 00:00:00	3160	615	64	58549.49	Cancelled	Yes
15	2022-10-08 00:00:00	3074	791	102	35850.73	Active	Yes
16	2023-10-18 00:00:00	4649	738	101	24103.26	Active	No
17	2022-08-17 00:00:00	1394	332	95	35096.43	Active	No
18	2024-05-25 00:00:00	4933	530	125	8528.06	Lapsed	Yes
19	2022-11-13 00:00:00	2336	479	60	29712.91	Active	No
20	2023-07-25 00:00:00	1696	581	86	40668.67	Active	Yes
21	2023-05-23 00:00:00	4556	481	125	54519.06	Active	Yes
22	2023-10-12 00:00:00	4761	319	143	25600.2	Active	Yes
23	2022-04-04 00:00:00	3932	355	65	48832.2	Lapsed	No
24	2022-03-28 00:00:00	1903	301	130	53071.45	Lapsed	No
25	2023-01-22 00:00:00	4851	691	141	25655.68	Lapsed	No
26	2023-10-30 00:00:00	4324	278	117	18163.11	Active	No
27	2024-05-26 00:00:00	4614	556	90	35738.36	Active	Yes
28	2023-02-20 00:00:00	1570	385	104	10763.5	Active	Yes
29	2024-06-16 00:00:00	4486	474	78	23226.98	Active	No

Customer Data:

This table contains the demographic profile of the client base. It is essential for segmenting customers to find cross-sell targets.

- **CustomerID:** The unique identifier for each customer (Primary Key).
- **CustomerName:** The name of the individual.
- **Gender / Age:** Demographic filters used to identify life-stage needs (e.g., younger customers might need travel insurance; older might need life insurance).
- **City:** Geographical location, useful for regional trend analysis.
- **IncomeSegment:** Categorizes customers (High, Medium, Low). High-income segments are often prime targets for premium cross-sell products like Wealth or Life insurance.

CustomerID	CustomerName	Gender	Age	City	IncomeSegment
3661	Customer_3661	Male	56	Chennai	High
4100	Customer_4100	Male	52	Delhi	Low
2830	Customer_2830	Female	57	Pune	Medium
3865	Customer_3865	Female	46	Pune	Low
3141	Customer_3141	Male	40	Bangalore	Low
1549	Customer_1549	Male	24	Bangalore	Medium
3950	Customer_3950	Male	25	Hyderabad	High
2238	Customer_2238	Male	27	Mumbai	High
3157	Customer_3157	Female	51	Hyderabad	High
4367	Customer_4367	Female	63	Bangalore	High
3459	Customer_3459	Male	51	Delhi	High
1070	Customer_1070	Male	50	Bangalore	High
1733	Customer_1733	Male	58	Delhi	Medium
3160	Customer_3160	Female	29	Chennai	Low
3074	Customer_3074	Male	21	Hyderabad	Medium
4649	Customer_4649	Female	32	Delhi	Low
1394	Customer_1394	Male	43	Mumbai	High
4933	Customer_4933	Male	43	Delhi	High
2336	Customer_2336	Female	30	Bangalore	Low
1696	Customer_1696	Male	27	Pune	High
4556	Customer_4556	Female	35	Hyderabad	High
4761	Customer_4761	Male	36	Hyderabad	Low
3932	Customer_3932	Male	32	Bangalore	Medium
1903	Customer_1903	Female	21	Delhi	High
4851	Customer_4851	Female	31	Pune	High
4324	Customer_4324	Female	53	Delhi	High
4614	Customer_4614	Female	52	Bangalore	Low
1570	Customer_1570	Male	23	Bangalore	Low
4486	Customer_4486	Male	28	Hyderabad	High

Policies Details:

This is our product catalog. It defines what you are selling.

- **PolicyID:** Unique identifier for each policy type.

- **PolicyName:** The specific name of the insurance product.
- **PolicyType:** The category (e.g., **Life, Health, Motor, Travel, Home**). This is the most critical column for "White Space" analysis (identifying which types a customer *doesn't* have).
- **PolicyCategory:** Defines if it is an Individual, Family, or Corporate plan.
- **RiskLevel:** The risk rating associated with the policy.

PolicyID	PolicyName	PolicyType	PolicyCategory	RiskLevel
491	Policy_491	Travel	Individual	Medium
546	Policy_546	Life	Family	Medium
794	Policy_794	Travel	Family	Medium
223	Policy_223	Travel	Individual	Low
486	Policy_486	Home	Corporate	Low
765	Policy_765	Home	Family	Low
753	Policy_753	Auto	Family	Low
705	Policy_705	Health	Family	High
629	Policy_629	Health	Corporate	High
256	Policy_256	Auto	Family	Low
597	Policy_597	Auto	Corporate	Low
596	Policy_596	Auto	Individual	Medium
615	Policy_615	Auto	Individual	High
791	Policy_791	Travel	Corporate	Medium
738	Policy_738	Auto	Individual	Low
332	Policy_332	Health	Corporate	Low
530	Policy_530	Life	Individual	Low
479	Policy_479	Health	Family	Medium
581	Policy_581	Life	Individual	Medium
481	Policy_481	Life	Individual	Low
319	Policy_319	Life	Corporate	High
355	Policy_355	Life	Individual	Low
301	Policy_301	Health	Corporate	Low
691	Policy_691	Life	Corporate	High
278	Policy_278	Health	Individual	Medium
556	Policy_556	Auto	Individual	High
385	Policy_385	Travel	Individual	Medium
474	Policy_474	Health	Family	High
310	Policy_310	Auto	Individual	Medium

Agents:

This table describes the sales force and distribution network.

- **AgentID:** Unique identifier for the agent.
- **AgentName:** The name of the agent responsible for the sale.
- **Channel:** How the sale was made (**Online, Branch, Call Center**). This helps you see if certain channels are better at cross-selling than others.
- **Region:** The broad territory (North, South, East, West) the agent operates in.

AgentID	AgentName	Channel	Region
95	Agent_95	Call Center	South
57	Agent_57	Call Center	West
127	Agent_127	Branch	East
118	Agent_118	Online	East
93	Agent_93	Call Center	North
114	Agent_114	Call Center	East
139	Agent_139	Branch	South
122	Agent_122	Branch	West
67	Agent_67	Agent	West
79	Agent_79	Agent	North
136	Agent_136	Branch	North
101	Agent_101	Online	South
71	Agent_71	Branch	South
64	Agent_64	Online	North
102	Agent_102	Branch	South
125	Agent_125	Branch	West
60	Agent_60	Agent	North
86	Agent_86	Online	North
143	Agent_143	Branch	East
65	Agent_65	Online	West
130	Agent_130	Agent	West
141	Agent_141	Branch	East
117	Agent_117	Online	North
90	Agent_90	Agent	South
104	Agent_104	Branch	North
78	Agent_78	Branch	East
142	Agent_142	Call Center	West
73	Agent_73	Call Center	West
134	Agent_134	Online	East

Payment Details:

This adds a layer of behavioral data regarding how customers manage their finances.

- **PaymentID:** Unique record for the payment.
- **PolicyTransactionID:** Links the payment back to the specific sale.
- **PaymentMode:** How they pay (**Debit Card, UPI, Credit Card, Net Banking**).
- **PaymentFrequency:** How often they pay (**Monthly, Quarterly, Yearly**). Yearly payers are often more stable and might be better targets for long-term cross-sell products

PaymentID	PolicyTransactionID	PaymentMode	PaymentFrequency
1	1	Debit Card	Monthly
2	2	UPI	Yearly
3	3	UPI	Yearly
4	4	UPI	Monthly
5	5	Debit Card	Quarterly
6	6	Credit Card	Monthly
7	7	Credit Card	Yearly
8	8	UPI	Half-Yearly
9	9	Credit Card	Quarterly
10	10	Debit Card	Yearly
11	11	UPI	Yearly
12	12	Net Banking	Quarterly
13	13	Credit Card	Monthly
14	14	Credit Card	Yearly
15	15	Debit Card	Monthly
16	16	Credit Card	Monthly
17	17	Debit Card	Monthly
18	18	UPI	Quarterly
19	19	Debit Card	Quarterly
20	20	Net Banking	Quarterly
21	21	UPI	Half-Yearly
22	22	Debit Card	Quarterly
23	23	Credit Card	Quarterly
24	24	Credit Card	Half-Yearly
25	25	Net Banking	Quarterly
26	26	Net Banking	Half-Yearly
27	27	Net Banking	Yearly
28	28	Net Banking	Quarterly
29	29	Net Banking	Quarterly
30	30	Debit Card	Half-Yearly
31	31	Debit Card	Yearly

Data Cleaning & Preparation

- Steps performed:
- Removed cancelled policies
- Handled missing and duplicate IDs
- Removed duplicate agent IDs
- Standardized text fields
- Created a CrossSell Helper flag for easy use in arithmetic operations.
- Created derived columns (Year, Month, Age Group)

Policy Transactions after cleanup and preprocessing:

PolicyTransactionID	TransactionDate	CustomerID	PolicyID	AgentID	PremiumAmount	PolicyStatus	CrossSellFlag	Cross Sell Helper	Year	Month	Month Number	Clean PolicyStatus	Clean Cross Flag
1	2023-06-05 00:00:00	3661	491	95	22730.74	Lapsed	No	0	2023	Jun	6	Lapsed	No
2	2024-03-03 00:00:00	4100	546	57	38343.47	Active	No	0	2024	Mar	3	Active	No
3	2024-04-15 00:00:00	2830	794	127	58163.51	Active	No	0	2024	Apr	4	Active	No
4	2024-05-21 00:00:00	3865	273	118	41132.6	Active	No	0	2024	May	5	Active	No
5	2024-05-05 00:00:00	3141	486	93	40539.86	Active	Yes	1	2024	May	5	Active	Yes
6	2022-03-21 00:00:00	1549	765	114	29286.8	Active	Yes	1	2022	Mar	3	Active	Yes
7	2022-12-17 00:00:00	3950	753	139	11498.24	Active	Yes	1	2022	Dec	12	Active	Yes
8	2024-05-16 00:00:00	2238	705	122	16857.51	Lapsed	Yes	1	2024	May	5	Lapsed	Yes
9	2022-02-23 00:00:00	3157	629	67	14215.99	Lapsed	No	0	2022	Feb	2	Lapsed	No
10	2023-08-02 00:00:00	4367	256	79	24054.32	Active	No	0	2023	Aug	8	Active	No
11	2024-01-09 00:00:00	3459	753	136	27463.44	Active	No	0	2024	Jan	1	Active	No
12	2023-06-11 00:00:00	1070	597	101	14421.87	Active	No	0	2023	Jun	6	Active	No
15	2022-10-08 00:00:00	3074	791	102	35850.73	Active	Yes	1	2022	Oct	10	Active	Yes
16	2023-10-18 00:00:00	4649	738	101	24103.26	Active	No	0	2023	Oct	10	Active	No
17	2022-08-17 00:00:00	1394	332	95	35096.43	Active	No	0	2022	Aug	8	Active	No
18	2024-05-25 00:00:00	4933	530	125	8528.06	Lapsed	Yes	1	2024	May	5	Lapsed	Yes
19	2022-11-13 00:00:00	2336	479	60	29712.91	Active	No	0	2022	Nov	11	Active	No
20	2023-07-25 00:00:00	1696	581	86	40668.67	Active	Yes	1	2023	Jul	7	Active	Yes
21	2023-05-23 00:00:00	4556	481	125	54519.06	Active	Yes	1	2023	May	5	Active	Yes
22	2023-10-12 00:00:00	4761	319	143	25600.2	Active	Yes	1	2023	Oct	10	Active	Yes
23	2022-04-04 00:00:00	3932	355	65	48832.2	Lapsed	No	0	2022	Apr	4	Lapsed	No
24	2022-03-28 00:00:00	1903	301	130	53071.45	Lapsed	No	0	2022	Mar	3	Lapsed	No
25	2023-01-22 00:00:00	4851	691	141	25655.68	Lapsed	No	0	2023	Jan	1	Lapsed	No
26	2023-10-30 00:00:00	4324	278	117	18163.11	Active	No	0	2023	Oct	10	Active	No
27	2024-05-26 00:00:00	4614	556	90	35738.36	Active	Yes	1	2024	May	5	Active	Yes
28	2023-02-20 00:00:00	1570	385	104	10763.5	Active	Yes	1	2023	Feb	2	Active	Yes
29	2024-06-16 00:00:00	4486	474	78	23226.98	Active	No	0	2024	Jun	6	Active	No
30	2022-05-22 00:00:00	3349	310	57	22767.17	Lapsed	No	0	2022	May	5	Lapsed	No
31	2024-01-28 00:00:00	2065	350	142	7463.91	Active	Yes	1	2024	Jan	1	Active	Yes
32	2023-12-25 00:00:00	3264	560	67	24438.57	Active	No	0	2023	Dec	12	Active	No
33	2023-09-05 00:00:00	3314	796	73	3808.3	Active	No	0	2023	Sep	9	Active	No
34	2022-01-05 00:00:00	1043	262	86	12131.17	Active	Yes	1	2022	Jan	1	Active	Yes
35	2023-08-27 00:00:00	4814	438	134	24018.02	Active	Yes	1	2023	Aug	8	Active	Yes
36	2024-04-15 00:00:00	3868	268	87	58985.75	Active	Yes	1	2024	Apr	4	Active	Yes

Customer Data after cleanup and preprocessing:

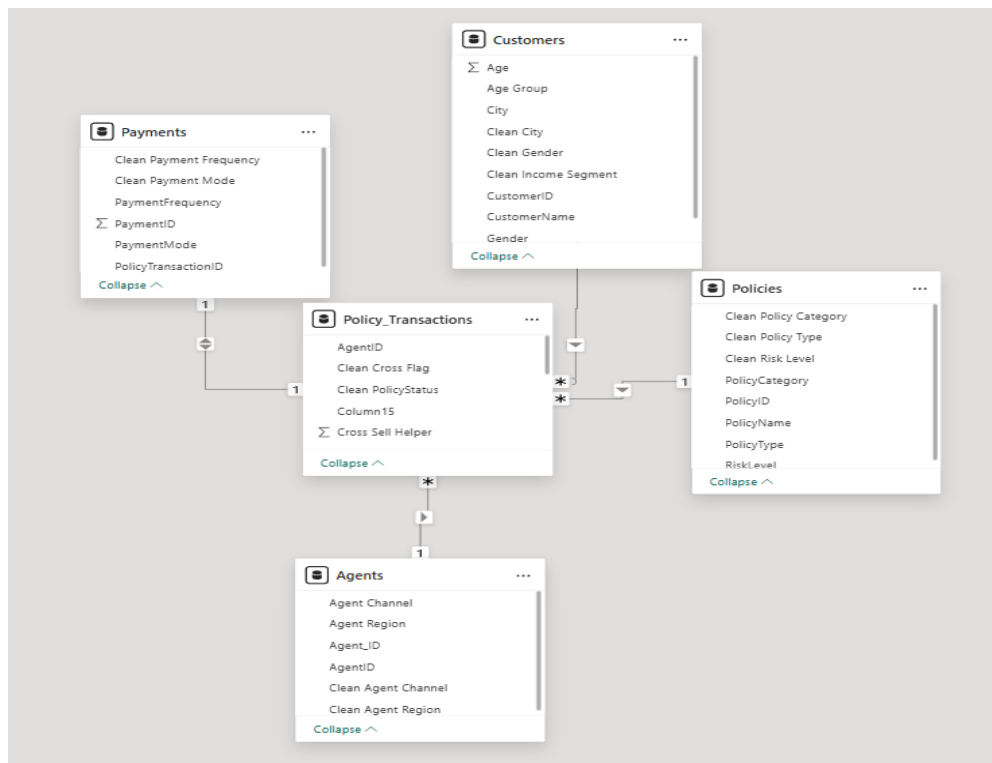
CustomerID	CustomerName	Gender	Age	City	IncomeSegment	Clean City	Clean Income Segment	Clean Gender	Age Group
3661	Customer_3661	Male	56	Chennai	High	Chennai	High	Male	46-60
4100	Customer_4100	Male	52	Delhi	Low	Delhi	Low	Male	46-60
2830	Customer_2830	Female	57	Pune	Medium	Pune	Medium	Female	46-60
3865	Customer_3865	Female	46	Pune	Low	Pune	Low	Female	46-60
3141	Customer_3141	Male	40	Bangalore	Low	Bangalore	Low	Male	36-45
1549	Customer_1549	Male	24	Bangalore	Medium	Bangalore	Medium	Male	18-25
3950	Customer_3950	Male	25	Hyderabad	High	Hyderabad	High	Male	18-25
2238	Customer_2238	Male	27	Mumbai	High	Mumbai	High	Male	26-35
3157	Customer_3157	Female	51	Hyderabad	High	Hyderabad	High	Female	46-60
4367	Customer_4367	Female	63	Bangalore	High	Bangalore	High	Female	60+
3459	Customer_3459	Male	51	Delhi	High	Delhi	High	Male	46-60
1070	Customer_1070	Male	50	Bangalore	High	Bangalore	High	Male	46-60
1733	Customer_1733	Male	58	Delhi	Medium	Delhi	Medium	Male	46-60
3160	Customer_3160	Female	29	Chennai	Low	Chennai	Low	Female	26-35
3074	Customer_3074	Male	21	Hyderabad	Medium	Hyderabad	Medium	Male	18-25
4649	Customer_4649	Female	32	Delhi	Low	Delhi	Low	Female	26-35
1394	Customer_1394	Male	43	Mumbai	High	Mumbai	High	Male	36-45
4933	Customer_4933	Male	43	Delhi	High	Delhi	High	Male	36-45
2336	Customer_2336	Female	30	Bangalore	Low	Bangalore	Low	Female	26-35
1696	Customer_1696	Male	27	Pune	High	Pune	High	Male	26-35
4556	Customer_4556	Female	35	Hyderabad	High	Hyderabad	High	Female	26-35
4761	Customer_4761	Male	36	Hyderabad	Low	Hyderabad	Low	Male	36-45
3932	Customer_3932	Male	32	Bangalore	Medium	Bangalore	Medium	Male	26-35
1903	Customer_1903	Female	21	Delhi	High	Delhi	High	Female	18-25
4851	Customer_4851	Female	31	Pune	High	Pune	High	Female	26-35
4324	Customer_4324	Female	53	Delhi	High	Delhi	High	Female	46-60
4614	Customer_4614	Female	52	Bangalore	Low	Bangalore	Low	Female	46-60
1570	Customer_1570	Male	23	Bangalore	Low	Bangalore	Low	Male	18-25
4486	Customer_4486	Male	28	Hyderabad	High	Hyderabad	High	Male	26-35
3349	Customer_3349	Male	37	Hyderabad	High	Hyderabad	High	Male	36-45
2065	Customer_2065	Male	44	Pune	Low	Pune	Low	Male	36-45

Policy Data after cleanup and preprocessing:

PolicyID	PolicyName	PolicyType	PolicyCategory	RiskLevel	Clean Policy Type	Clean Policy Category	Clean Risk Level
491	Policy_491	Travel	Individual	Medium	Travel	Individual	Medium
546	Policy_546	Life	Family	Medium	Life	Family	Medium
794	Policy_794	Travel	Family	Medium	Travel	Family	Medium
223	Policy_223	Travel	Individual	Low	Travel	Individual	Low
486	Policy_486	Home	Corporate	Low	Home	Corporate	Low
765	Policy_765	Home	Family	Low	Home	Family	Low
753	Policy_753	Auto	Family	Low	Auto	Family	Low
705	Policy_705	Health	Family	High	Health	Family	High
629	Policy_629	Health	Corporate	High	Health	Corporate	High
256	Policy_256	Auto	Family	Low	Auto	Family	Low
597	Policy_597	Auto	Corporate	Low	Auto	Corporate	Low
596	Policy_596	Auto	Individual	Medium	Auto	Individual	Medium
615	Policy_615	Auto	Individual	High	Auto	Individual	High
791	Policy_791	Travel	Corporate	Medium	Travel	Corporate	Medium
738	Policy_738	Auto	Individual	Low	Auto	Individual	Low
332	Policy_332	Health	Corporate	Low	Health	Corporate	Low
530	Policy_530	Life	Individual	Low	Life	Individual	Low
479	Policy_479	Health	Family	Medium	Health	Family	Medium
581	Policy_581	Life	Individual	Medium	Life	Individual	Medium
481	Policy_481	Life	Individual	Low	Life	Individual	Low
319	Policy_319	Life	Corporate	High	Life	Corporate	High
355	Policy_355	Life	Individual	Low	Life	Individual	Low
301	Policy_301	Health	Corporate	Low	Health	Corporate	Low
691	Policy_691	Life	Corporate	High	Life	Corporate	High
278	Policy_278	Health	Individual	Medium	Health	Individual	Medium
556	Policy_556	Auto	Individual	High	Auto	Individual	High
385	Policy_385	Travel	Individual	Medium	Travel	Individual	Medium
474	Policy_474	Health	Family	High	Health	Family	High
310	Policy_310	Auto	Individual	Medium	Auto	Individual	Medium
350	Policy_350	Health	Corporate	Low	Health	Corporate	Low
560	Policy_560	Home	Family	High	Home	Family	High
707	Policy_707	Travel	Individual	Low	Travel	Individual	Low

Data Modeling

A star schema was implemented with Policy_Transactions as the fact table and Customers, Policies, Agents, Payments, and Date as dimensions.



DAX Measures

Key measures created:

- Cross Sell Percentage
- Cross Sell Premium

Formulas used for these calculations are as below.

Cross Sell Percentage = $\text{DIVIDE}(\text{sum}(\text{Policy_Transactions}[\text{Cross Sell Helper}]), \text{COUNT}(\text{Payments}[\text{PolicyTransactionID}])) * 100$

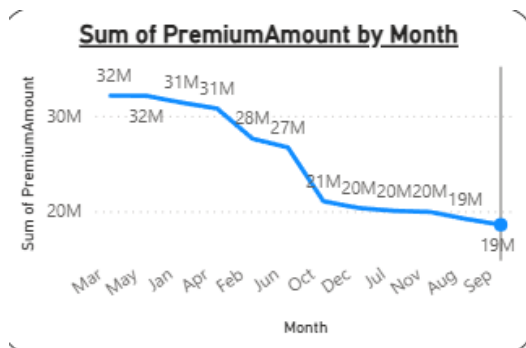
Cross Sell Premium = $\text{CALCULATE}(\text{SUM}(\text{Policy_Transactions}[\text{PremiumAmount}]), \text{Policy_Transactions}[\text{CrossSellFlag}] = \text{"Yes"})$

Dashboard Design

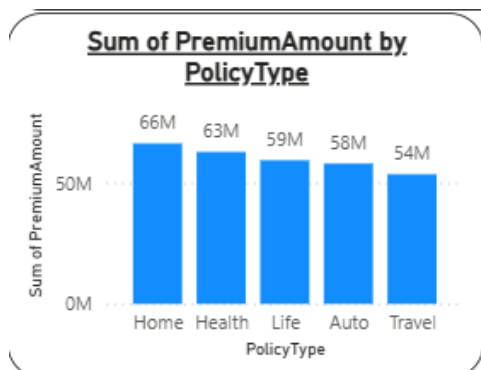
The dashboard includes KPI cards, monthly premium trends, cross-sell analysis by channel and policy, and interactive slicers.

Below are the list of interactive dashboards created.

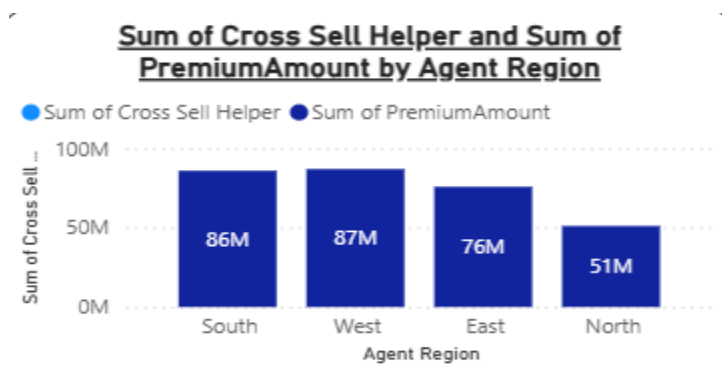
A Line chart was used to display monthly trends of premium amounts received.



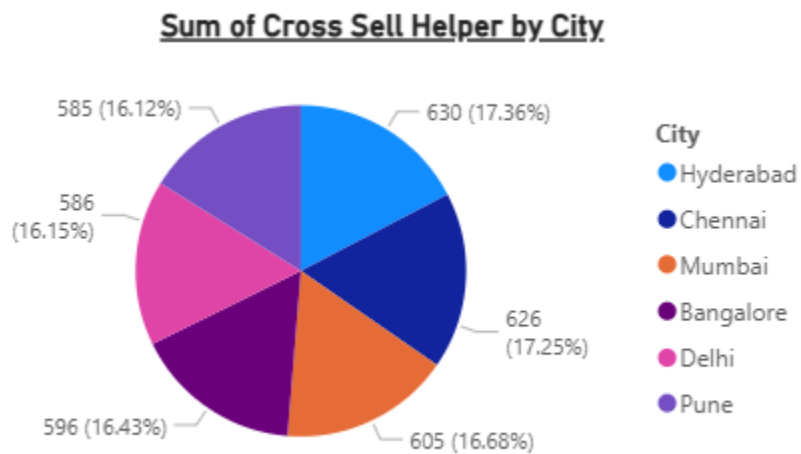
A Column chart was used to display policy amount received based on insurance policy type.



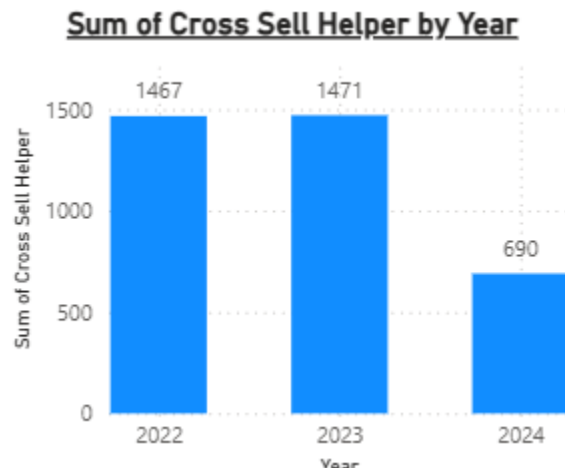
A column chart was used to display region wise total premium amount



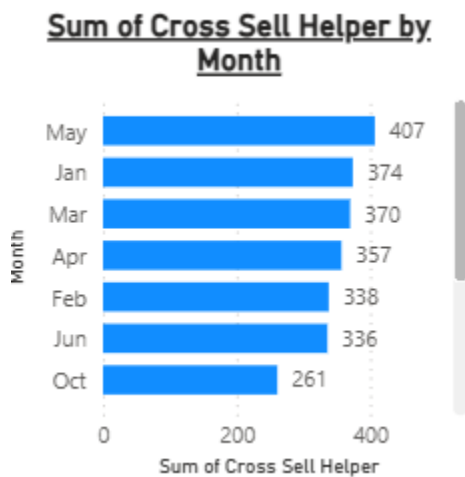
A pie chart was used to display City wise Cross selling percentage



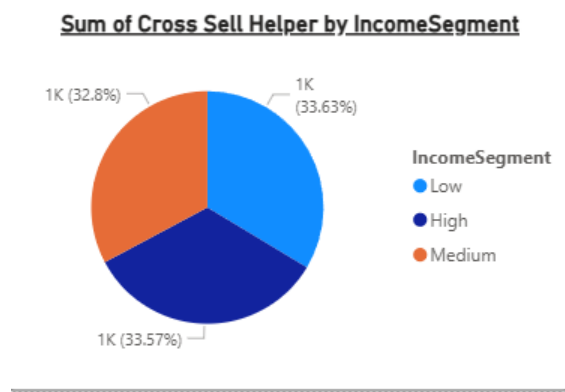
A Column chart was used to display year wise cross sell count



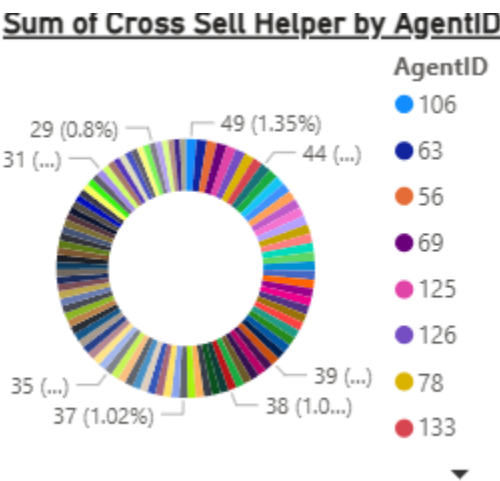
A Bar chart was used to display month wise cross cell count



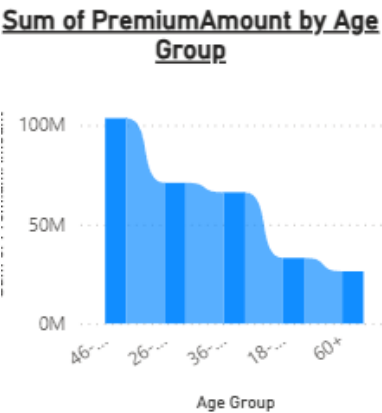
A pie chart used to display customer income segment



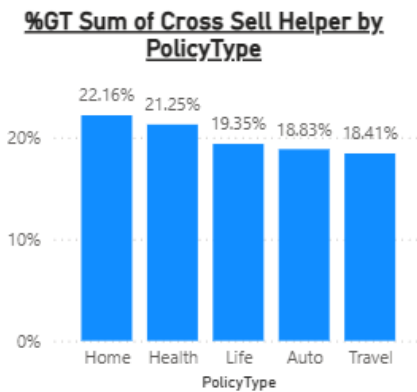
A Donut chart used to display Agent wise cross sell



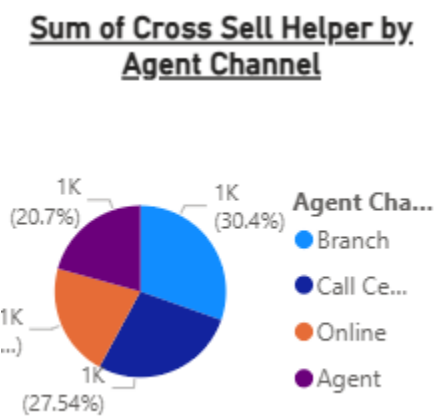
A ribbon chart used to display age wise premium amount



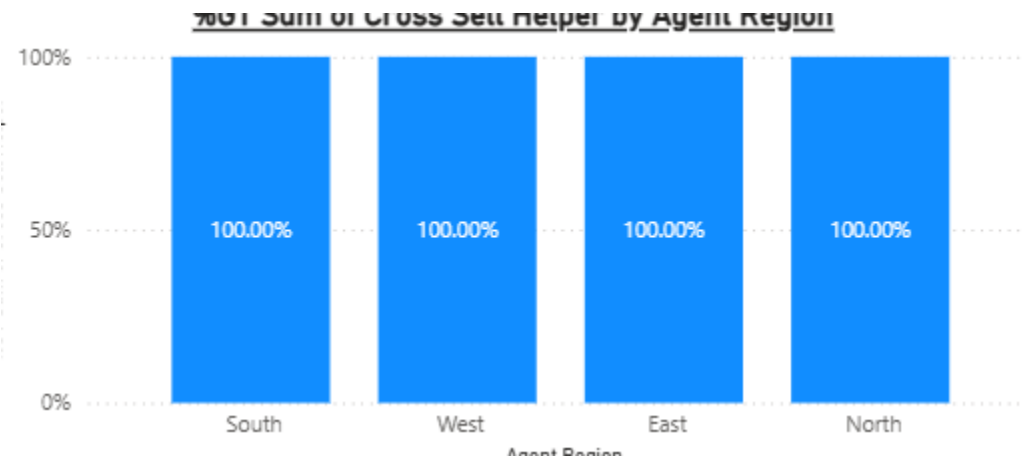
A column chart used to display year wise policy crosscell percentage



A pie chart used to display Agent cross cell percentage region wise



A column chart used to display region wise cross cell percentage



Key Insights

- Health and Life policies generate the highest premium
- Agent channel performs best for cross-sell
- High-income customers show higher acceptance

Business Recommendations

- Focus on high-performing policies which are Home and Health Insurance.
- Strengthen agent channel incentives. Branch sales have highest CrossSell percentage.
- Target high-value customer segments. Customers with highest cross sell percentage are customers with home insurance, aged above 45 with low- and high-income brackets.

Expected Outcomes

- Improved visibility into cross-sell performance
- Data-driven decision making
- Reusable reporting framework

Limitations

- Historical data only
- No predictive analytics
- Dependent on data quality

Conclusion

This project demonstrates an end-to-end analytics solution using Excel and Power BI, transforming raw insurance data into actionable insights.